

## CHARTERS CREEK, South Africa

WMO: 684910

Lat: 28.1978S

Lon: 32.4142E

Elev: 8

StdP: 101.23

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 8.9        | 10.4 | 3.5                           | 4.8 | 15.6 | 6.5 | 6.0 | 14.4 | 7.1                   | 20.3 | 6.4 | 21.9 | 0.8                   | 260  | 0.298 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 7.3                    | 31.7            | 25.0 | 30.6 | 24.6 | 29.7 | 24.3 | 26.8                | 29.5 | 26.0 | 29.1 | 25.4 | 28.4 | 5.0                  | 40   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.2                            | 21.6 | 28.0 | 25.2 | 20.3 | 27.6 | 24.5 | 19.5 | 27.2 | 84.2          | 29.7 | 80.7 | 29.1 | 77.9 | 28.5 | 30.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 7.3               | 6.4  | 5.9 | 5.6                        | 36.0 | 1.4                | 1.6 | 4.6                                                | 37.1 | 3.7        | 38.1 | 2.9        | 38.9 | 1.9        | 40.1 |    |    |
|                   |      |     | 4.7                        | 27.6 | 1.6                | 0.8 | 3.5                                                | 28.2 | 2.6        | 28.7 | 1.7        | 29.2 | 0.6        | 29.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 21.7   | 24.8  | 25.0  | 24.7  | 22.4  | 20.5  | 18.3  | 18.0  | 19.2  | 20.3  | 20.9  | 22.4  | 24.0  |  |  |
|                                                                        | DBStd       | 3.37   | 2.07  | 1.99  | 2.05  | 2.41  | 2.08  | 2.17  | 2.25  | 2.53  | 2.64  | 2.66  | 2.55  | 2.44  |  |  |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | HDD18.3     | 108    | 0     | 0     | 0     | 2     | 5     | 26    | 34    | 20    | 11    | 8     | 2     | 1     |  |  |
|                                                                        | CDD10.0     | 4266   | 459   | 420   | 454   | 373   | 327   | 251   | 247   | 284   | 308   | 339   | 372   | 433   |  |  |
|                                                                        | CDD18.3     | 1333   | 201   | 187   | 196   | 124   | 74    | 27    | 22    | 46    | 69    | 88    | 124   | 175   |  |  |
|                                                                        | CDH23.3     | 7717   | 1508  | 1372  | 1321  | 586   | 256   | 93    | 79    | 136   | 258   | 351   | 603   | 1154  |  |  |
|                                                                        | CDH26.7     | 1712   | 397   | 345   | 318   | 102   | 23    | 8     | 8     | 19    | 49    | 56    | 107   | 281   |  |  |
| (14) Wind                                                              | WSAvg       | 2.8    | 2.8   | 2.8   | 2.7   | 2.5   | 2.3   | 2.2   | 2.3   | 2.9   | 3.2   | 3.3   | 3.2   | 3.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 899    | 118   | 111   | 108   | 65    | 46    | 34    | 32    | 34    | 50    | 107   | 102   | 91    |  |  |
|                                                                        | PrecMax     | 1566   | 269   | 283   | 262   | 172   | 314   | 138   | 121   | 151   | 203   | 326   | 333   | 209   |  |  |
|                                                                        | PrecMin     | 421    | 16    | 22    | 18    | 14    | 0     | 0     | 0     | 0     | 3     | 22    | 20    | 27    |  |  |
|                                                                        | PrecStd     | 260    | 65    | 59    | 66    | 37    | 62    | 37    | 34    | 34    | 46    | 72    | 74    | 48    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 32.5  | 32.9  | 32.3  | 30.8  | 28.7  | 27.6  | 27.8  | 28.8  | 30.7  | 30.5  | 31.2  | 33.0  |  |  |
|                                                                        |             | MCWB   | 25.3  | 25.4  | 25.8  | 24.0  | 21.2  | 18.4  | 17.9  | 18.9  | 21.2  | 23.2  | 24.6  | 25.6  |  |  |
|                                                                        | 2%          | DB     | 31.0  | 31.0  | 30.7  | 29.0  | 27.2  | 25.6  | 25.3  | 26.2  | 27.6  | 28.0  | 29.1  | 30.8  |  |  |
|                                                                        |             | MCWB   | 24.7  | 25.1  | 25.4  | 23.5  | 21.2  | 18.7  | 18.3  | 19.7  | 21.1  | 22.1  | 23.5  | 25.0  |  |  |
|                                                                        | 5%          | DB     | 29.9  | 29.8  | 29.6  | 27.7  | 26.0  | 24.2  | 23.9  | 24.6  | 25.9  | 26.6  | 27.7  | 29.4  |  |  |
|                                                                        |             | MCWB   | 24.2  | 24.8  | 24.9  | 23.0  | 20.7  | 18.4  | 18.2  | 19.2  | 20.3  | 21.5  | 23.0  | 24.4  |  |  |
|                                                                        | 10%         | DB     | 28.9  | 28.9  | 28.5  | 26.5  | 24.8  | 22.8  | 22.5  | 23.3  | 24.4  | 25.3  | 26.5  | 28.1  |  |  |
|                                                                        |             | MCWB   | 23.9  | 24.4  | 24.3  | 22.3  | 20.2  | 18.1  | 17.8  | 18.5  | 19.7  | 20.8  | 22.4  | 23.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 27.4  | 27.9  | 27.8  | 25.6  | 23.0  | 21.7  | 21.2  | 22.2  | 22.8  | 24.1  | 25.4  | 26.7  |  |  |
|                                                                        |             | MCDB   | 29.6  | 29.7  | 29.7  | 28.2  | 26.8  | 24.4  | 23.8  | 25.8  | 28.5  | 28.9  | 29.5  | 30.9  |  |  |
|                                                                        | 2%          | WB     | 26.1  | 26.6  | 26.7  | 24.4  | 22.1  | 20.5  | 19.9  | 20.8  | 21.6  | 23.0  | 24.4  | 25.9  |  |  |
|                                                                        |             | MCDB   | 29.3  | 29.0  | 29.1  | 27.6  | 25.7  | 23.5  | 23.0  | 24.2  | 26.5  | 26.8  | 27.9  | 29.8  |  |  |
|                                                                        | 5%          | WB     | 25.3  | 25.8  | 25.7  | 23.6  | 21.4  | 19.6  | 19.1  | 20.0  | 20.9  | 22.1  | 23.7  | 25.0  |  |  |
|                                                                        |             | MCDB   | 28.5  | 28.6  | 28.3  | 26.7  | 24.7  | 22.6  | 22.3  | 23.3  | 25.1  | 25.5  | 26.8  | 28.4  |  |  |
|                                                                        | 10%         | WB     | 24.7  | 25.2  | 24.8  | 22.9  | 20.7  | 18.8  | 18.3  | 19.2  | 20.2  | 21.3  | 22.9  | 24.2  |  |  |
|                                                                        |             | MCDB   | 27.8  | 28.0  | 27.5  | 25.7  | 23.8  | 21.6  | 21.5  | 22.4  | 23.6  | 24.5  | 25.8  | 27.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 7.3   | 7.3   | 7.7   | 8.1   | 9.4   | 10.8  | 10.4  | 9.2   | 8.5   | 7.1   | 6.8   | 7.2   |  |  |
|                                                                        |             | MCDBR  | 9.0   | 8.6   | 8.8   | 9.0   | 10.5  | 12.2  | 12.2  | 10.6  | 10.3  | 9.1   | 8.7   | 9.2   |  |  |
|                                                                        |             | MCWBR  | 4.0   | 3.9   | 4.2   | 4.4   | 5.3   | 6.4   | 6.2   | 5.3   | 5.1   | 4.7   | 4.4   | 4.4   |  |  |
|                                                                        | 5% WB       | MCDBR  | 7.9   | 7.7   | 8.1   | 8.3   | 9.4   | 10.3  | 10.3  | 9.3   | 9.6   | 8.4   | 7.7   | 8.5   |  |  |
|                                                                        |             | MCWBR  | 4.3   | 4.0   | 4.5   | 4.4   | 5.4   | 6.5   | 6.1   | 5.1   | 5.0   | 4.6   | 4.3   | 4.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.446 | 0.430 | 0.430 | 0.411 | 0.384 | 0.371 | 0.376 | 0.441 | 0.493 | 0.513 | 0.479 | 0.465 |  |  |
|                                                                        | taud        |        | 2.339 | 2.389 | 2.377 | 2.394 | 2.435 | 2.448 | 2.428 | 2.223 | 2.096 | 2.079 | 2.210 | 2.267 |  |  |
|                                                                        | Ebn at Noon |        | 900   | 897   | 865   | 830   | 806   | 795   | 806   | 785   | 792   | 814   | 864   | 886   |  |  |
|                                                                        | Edh at Noon |        | 135   | 125   | 120   | 108   | 94    | 88    | 93    | 126   | 157   | 169   | 153   | 146   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 6.12  | 5.96  | 5.20  | 4.19  | 3.62  | 3.18  | 3.39  | 3.99  | 4.55  | 4.91  | 5.53  | 6.06  |  |  |
|                                                                        | RadStd      |        | 0.53  | 0.45  | 0.34  | 0.25  | 0.16  | 0.14  | 0.20  | 0.24  | 0.27  | 0.39  | 0.49  | 0.45  |  |  |

## Historical Trends

|                            | DBAvg | Heating |     | Cooling |       |       | Degree-Days |         |         |         |
|----------------------------|-------|---------|-----|---------|-------|-------|-------------|---------|---------|---------|
|                            |       | 99% DB  |     | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                            |       | (d)     | (e) | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only          | N/A   | N/A     | N/A | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page